

Microstructure Factor Zoo — AAPL

Executive Summary

MGMTMFE 412 — Market Frictions

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Objective

This project tests whether 11 real-time L2 order-book features derived from Databento MBP-10 data can generate profitable intraday trading signals for AAPL after accounting for bid-ask spread transaction costs. The central economic question is whether order-book predictability is already priced into the spread — consistent with the Glosten-Milgrom adverse-selection model of market making.

Data

Databento XNAS ITCH MBP-10 snapshots (10-level book) for AAPL across three non-overlapping periods:

Split	Dates	Days	1-second Bars
Training	Jan 2 – Feb 26, 2024	18	414,025
Validation	Feb 28 – Apr 1, 2024	7	161,088
Out-of-sample	Oct 1 – Oct 23, 2024	17	~391,000

The 6-month gap between validation (Q1 2024) and OOS (Oct 2024) provides a genuine regime-change test: Q1 was characterized by low-volatility consolidation, while October 2024 saw a post-iPhone-launch uptrend.

Prediction target: 60-second forward microprice return in basis points, winsorized at the 1st/99th percentile of training labels ($\sigma_y = 4.14$ bps).

Approach

Eleven stationary features were constructed from raw book data: four Order Flow Imbalance metrics (OFI at 1, 5, 30 seconds, and multilevel), four book-shape features (imbalance ratio, depth ratio, bid slope, ask slope), and three dynamic features (microprice drift over 10 seconds, spread in bps, realized volatility over 60 seconds).

Two models were trained: a Ridge regression (OLS, $\alpha = 1.0$) and a LightGBM ensemble (100 trees, max depth 4, early stopping). All features are normalized to zero mean and unit variance per trading day before prediction. Signals above $|\hat{r}_t| > 0.50$ bps trigger trades executed via a walk-book fill model against the live ask or bid side of the order book.

The break-even IC threshold is:

$$\text{IC}_{\text{break-even}} = \frac{\text{spread}_{\text{bps}}}{\sigma_y} = \frac{0.54}{4.14} = 0.130$$

Any model must achieve $\text{IC} \geq 0.130$ to generate positive expected PnL after paying the spread on both entry and exit legs.

Key Results

Model	Period	IC	Sharpe	Ann. Return	Hit Rate	N Trades	Avg Hold
OLS	Training	0.029	−7.1	−\$9,160	22.3%	260	23.5s
OLS	Validation	0.047	−9.8	−\$47,328	18.5%	275	21.3s
OLS	OOS Oct-24	0.012	−22.7	−\$86,225	19.5%	606	22.2s
Light-GBM	OOS Oct-24	−0.003	−3,956	−\$253,492	14.9%	2,680	17.4s

Hit rate = % of individual trades profitable. Avg Hold = actual mean holding period; target is 60s but 78% of trades exit early via stop-loss.

The OLS validation IC of 0.047 is statistically significant ($t \approx 18$, $p < 0.001$), confirming that L2 features do contain genuine predictive information about 60-second price moves. However, 0.047 is only **36% of the break-even threshold (0.130)** — the signal is real but far too weak to overcome transaction costs.

In October 2024 (OOS), the OLS IC collapses to 0.012 — a **76% attenuation** from validation. The primary cause is the **realized_vol** feature: it captures Q1 2024 volatility clustering, which does not transfer to the October momentum regime. Removing **realized_vol** in the ablation study drops training IC by 0.0114 (39% of total IC), making it the single most important and most brittle feature.

LightGBM overfits severely: training IC of 0.103 falls to −0.003 OOS, generating 2,680 trades and losing \$253,492 due to massive over-trading on noise signals.

The Break-Even IC Barrier

Every period’s IC falls well below the 0.130 break-even threshold:

Period	OLS IC	Break-Even IC	Ratio
Training	0.029	0.130	0.22×
Validation	0.047	0.130	0.36×
OOS Oct-24	0.012	0.130	0.09×

The AAPL bid-ask spread averages 0.54 bps. At AAPL’s price of approximately \$185, one tick (\$0.01) represents 5.4 bps — meaning even the minimum tick spread imposes a high bar. A taker strategy on AAPL must generate nearly 3× more IC than the best observed validation IC just to break even.

Risk Controls and Trade Mechanics

All trades follow strict risk rules: - **Stop-loss** at $1.5\times$ the entry spread from the entry midprice (not fill price) — triggered for 78% of OLS OOS trades - **Daily kill switch** halts trading if PnL falls below -2% of \$50,000 notional - **Inventory cap** at ± 500 shares prevents runaway accumulation - **Walk-book fill model** realistically accounts for book depth consumption at entry and exit

The stop-loss triggering at 78% of trades confirms that even the modest directional signal embedded in OFI and book imbalance features is insufficient to consistently move price in the predicted direction before the stop-loss is reached.

Capacity Analysis

AUM	Almgren-Chriss Market Impact	Assessment
\$10M	0.3 bps	Within signal edge (but strategy still loses)
\$100M	2.5 bps	Impact cost exceeds already-negative alpha
\$1B	25.4 bps	Completely consumes any conceivable edge

The strategy is capacity-constrained below \$10M and unprofitable even there.

CAPM Attribution

Period	β_{SPY}	α_{ann}
Training	-0.06	-0.17
Validation	+0.39	-0.89
OOS Oct-24	+0.46	-1.62

The strategy generates negative CAPM alpha across all periods. The positive OOS beta (+0.46) means the strategy takes inadvertent market exposure in October 2024 — it tends to buy on up-days (AAPL correlated with SPY), but the directional component only amplifies losses rather than generating systematic gain.

Conclusion

The signal is real but efficiently priced into the spread. AAPL’s L2 order-book features produce an IC of 0.047 on unseen validation data — statistically significant but economically insufficient, representing only 36% of the break-even threshold. The market maker has correctly set the half-spread equal to (or greater than) the expected 60-second adverse price move implied by these signals.

Three paths to profitability exist but none are achievable with this setup: (1) achieve $\text{IC} \geq 0.130$ through substantially better feature engineering or higher-frequency data, (2) become a market maker and collect rather than pay the spread, or (3) find a structural market inefficiency where the spread/volatility ratio is temporarily below equilibrium.

The project validates the Glosten-Milgrom model: in a liquid, informationally efficient market like AAPL, order-book predictability is already priced into the bid-ask spread, making taker strategies with predictive signals systematically unprofitable.